United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final Exam, Trimester: Spring 2026

Course Code: CSE 113/EEE 2113; Course Title: Electrical Circuits

Total Marks: 40; Duration: 2 hour

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. **(12 Marks)**

[6+4+2]
CO3

A **linear source network (Figure 1)** is accessible through its **output terminals A–B**. The network contains **an unknown internal resistance (R)**. An **experimental measurement** obtained by connecting a **load** across the terminals showed V_{AB} = 160 V while I_a = 3 A flowed into the load.

I. **Determine** the Thevenin equivalent voltage and resistance of the source network. (**Note:** Use the general approach as well as the data provided across the variable load while determining the Thevenin voltage and current)

II. Using the Thevenin resistance calculated above, **find** the value of **the internal resistance (R)**.

III. **Calculate** the maximum power that can be extracted from the network and **specify** the corresponding load resistance.

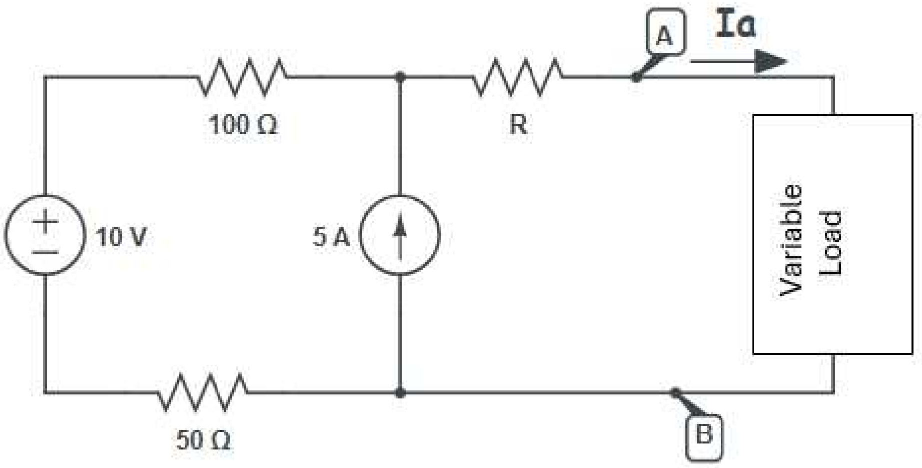

Figure 1

Question 2: Answer all the questions. **(8 Marks)**

[4+4]
CO3

I. For **Figure 2**, **find** the current, i, using the superposition theorem.

II. **Validate** the superposition theorem by finding the value of $\boldsymbol{i}$ without using superposition (you can use other methods except superposition theorem).

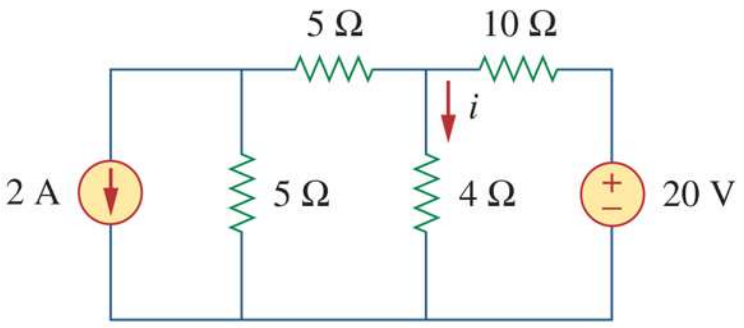

Figure 2

Question 3: Answer all the questions **(12 Marks)**

For **Figure 3**, **answer** the following questions: [6+2+2+2] **CO4**

I. **Determine** the **value** of Z_X and **the total impedance** of this circuit if $V_{L1} = 2\angle{-35}$ and $I_Z = 0.148\angle 20$.

II. **What** electric components (Resistors/Capacitors/Inductors) are used in Z_X, and **determine** their values.

III. **Determine** the source voltage's expression $V_1(t)$.

IV. **Explain** if this AC circuit's behaviour can be changed from inductive to a capacitive nature by changing the source's frequency. (**Note:** An AC circuit shows inductive behavior when its total impedance has positive reactance, and capacitive behavior when its total impedance has negative reactance.)

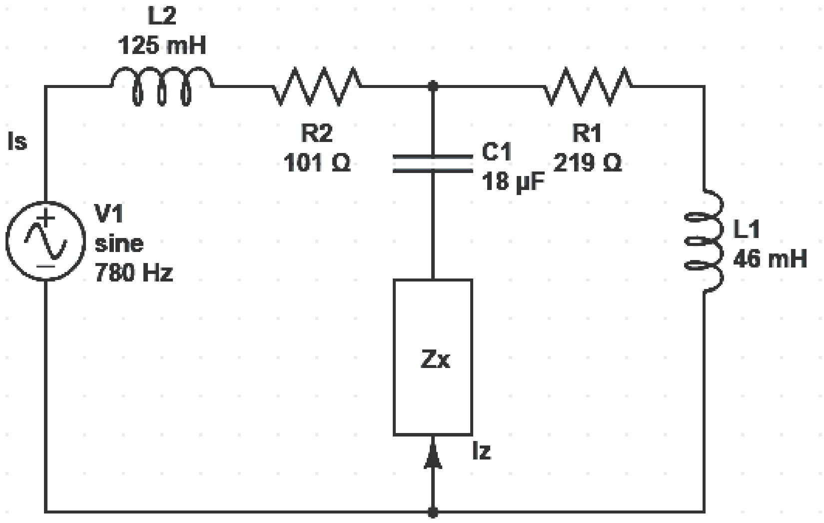

Figure 3

Question 4: Answer all the questions. **(8 Marks)**

A 10 Ω resistor receives a periodic voltage v(t) across it. The waveform is shown below **(Figure 4)**. Answer the following questions: [2+6] **CO4**

I. **Derive** the mathematical expression of v(t).

II. **Determine** the power absorbed by the resistor.

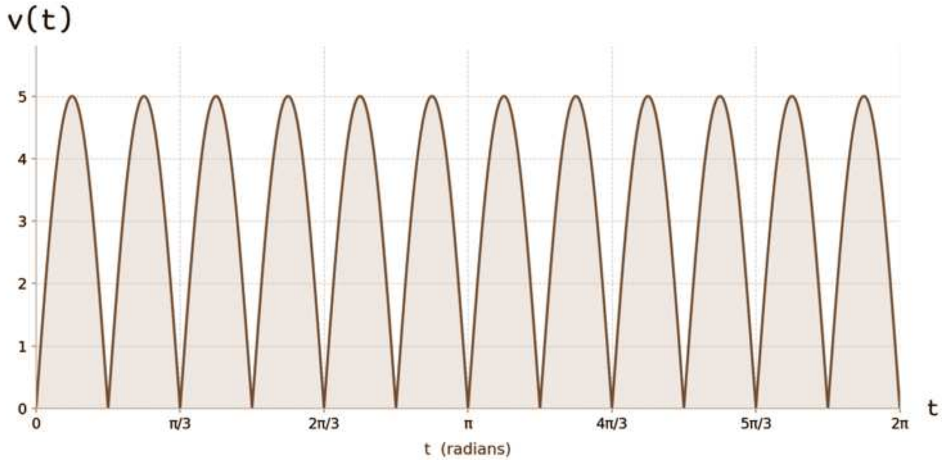

Figure 4